

Investigating Heart Disease Predictors in Survey-Based Categorical Data Through Machine Learning Models

Shrijan Subedi

School of Built Environment, Engineering, and Computing
Leeds Beckett University, Leeds, United Kingdom

Abstract—Global lifestyles have changed dramatically since industrialization began, and this has had a big impact on people’s health and wellness. Machine learning presents new chances to enhance heart health outcomes and assist medical professionals in diagnosing heart diseases. In this study, various machine learning and deep learning models have been evaluated in predicting heart disease using various behavioral and metabolic factors. A survey based dataset has been used to train the classifier. The findings conclude that Gradient Boosting algorithm performs the best while deep learning models like Feed-forward Neural Network and Convolutional Neural Networks have similar performance to Gradient Boosting model. The study also evaluates the various factors that impact an individual’s heart health by through a correlation test.

Index Terms—Heart health, Machine Learning, Convolutional Neural Network, Correlation, Gradient Boosting

I. INTRODUCTION

A. Background of Context

Global lifestyles have changed dramatically since industrialization began, and this has had a big impact on people’s health and wellness. Increased indulgence in harmful behaviors, poor diet, and sedentary habits have contributed to this trend, which has increased the prevalence of cardiovascular illnesses (CVDs). Due to a variety of risk factors, including dietary decisions, metabolic changes, and environmental factors, these heart and blood vessel illnesses cause a significant number of deaths worldwide [1]–[3].

Machine learning has emerged as a potential tool in healthcare in response to this trend, especially in the diagnosis and prediction of heart-related diseases. The emergence of machine learning has granted medical practitioners improved capacities to handle and evaluate large amounts of health information, allowing precise identification of diseases such as heart failure [4], [5]. Even with issues with standardization of algorithms [6], machine learning is still developing and presents new chances to enhance heart health outcomes. This study aims to investigate these variables as well as the ability of machine learning to improve heart health.

B. Aim and Research Objectives

The report has the following objectives:

- R01: Perform a literature review on:
 - Changing lifestyle due to modernization.

- Cardiovascular Diseases and risk factors.
- Role of machine learning in healthcare
- Diagnosing heart diseases with machine learning

- R02: Level 3 Research Objective: Perform machine learning analysis.
- R03: Level 4 Research Objective: Perform deep learning analysis.
- R04: Evaluate the performance of various machine learning models in predicting heart diseases using the dataset.
- R05: Identify the most influential parameters related to Heart Health using correlation
- R06: Provide recommendation based on analysis.

C. Rationale

Heart related diseases are a big problem in the modern world. There has been numerous research that propose machine learning algorithms to predict heart diseases. However, most studies use the same dataset with limited number of rows that cause the models to overfit the data. Evaluating just accuracy in such cases may not present a viable solution to predicting heart health. This study thus aims to take a different approach to predicting heart health.

D. Contribution of the Research

This study proposes a similar study of machine learning and deep learning algorithms but with a differently structured dataset. The data is highly categorical and contains large number of responses from survey participants. This can provide a different approach to predicting heart diseases by exploring the possibility of survey questions in creating a predictive model. Also, the study aims to evaluate the most influential factors when predicting heart health. The findings from this study can be further used to improve the effectiveness of machine learning models regarding this structure of dataset and allow individuals to take preventive measure towards the factors that influence heart health the most.

E. Organization of Report

The report is organized into the following sections:

- **Introduction:** This section presents the background, the rationale, and the objectives for the research.

- **Literature Review:** This section reviews the existing work done in predicting heart health through machine learning. Also, it explores the current behavioural changes leading to many heart diseases.
- **Methodology:** This section discusses the macro and the micro methodology involved in this research
- **Findings and discussion:** This section discusses the findings and results of the analysis performed.
- **Recommendations:** This section provides recommendation based on the findings of the analysis.
- **Conclusion and Future Work:** This section discusses how the research objectives were achieved and provides a summary of the findings

II. LITERATURE REVIEW

A. A Shift in global lifestyle

Our lifestyle has changed significantly due to industrialization, affecting the way we eat, work, move and access necessary amenities. On one side, healthcare is becoming more accessible due to the advent of mobile technologies and smart phones [1]. On the other side, however, industrialization has resulted in people seeking comfort and indulging in factors that degrade one's health.

Sedentary lifestyle is one of the major unhealthy behaviours emerging from industrialization. Whether at work, at home or during traversal, people seldom need to use their muscles or perform movements that challenge their cardiovascular system. A lack of physical activity might influence other factors such as obesity, high levels of LDL cholesterol and diabetes. Similarly, alcohol abuse and smoking are an increasing problem. Alcohol has been linked to cause cardio-depressive effects leading to problems such as heart rhythm disorder and, impairment of blood vessels. Smoking has similar negative effects, causing increased cholesterol levels and circular system diseases [2].

Nutrition is another overlooked aspect of modern society. Due to improper eating habits there is a growing number of people that are either overweight or obese or both. Lack of nutrition can severely hinder normal organism functioning leading to health issues such as diabetes, tumours, and circulatory system diseases [2].

B. Cardiovascular Diseases and risk factors

Cardiovascular diseases (CVD) are diseases that involve the heart, blood vessels, or obstruction of blood flow due to diseased vascular supply [3]. CVDs are a leading cause of death around the world. In 2011, it accounted for 30% of all deaths and 50% of deaths from non-communicable diseases. Ten years later, in 2021, the disease accounted for 20.5 million deaths globally [7].

There are several risk factors associated with CVDs. Those factors can be categorized into:

- **Behavioural factors:** This includes factors such as lack of physical activity, alcohol consumption, smoking and so on. These factors can be addressed by making changes to daily behaviours.

- **Metabolic factors:** These factors are influenced by inner cellular working of a human body. Factors such as high blood pressure, high levels of LDL cholesterol and diabetes fall under this factor.
- **Environmental factors:** External factors such as air pollution fall under this category.
- **Unmodifiable factors:** This category includes factors that cannot be changed by taking preventive measures or treatments. Family history can be attributed to being an unmodifiable factor.

Although, some factors such as family history cannot be changed or acted upon, most factors are modifiable with lifestyles changes or treatment [7]. These changes can have substantial improvement in heart health. Factors such as obesity, diabetes, smoking, and high blood pressure have been studied to increase the likelihood of heart related disease such as heart failure [8]. On the contrary, exercise has been found to be helpful in reducing these risk factors either directly or indirectly [9]. It is thus necessary to understand what factor among these can help the most in improving a person's cardiovascular health. Also, by evaluating these factors, professionals can assess if an individual has a risk of developing heart disease. For this reason, the factors mentioned have been selected for analysis in this research.

C. Role of artificial intelligence in modern healthcare

Artificial intelligence has become an important addition to many sectors. Even more, machine learning has become an important asset in healthcare sector due to the massive amount of health data generated worldwide [4]. The vast amount of data allows the machine learning models to detect diseases such as cancer, diabetes, Alzheimer's disease accurately [5]. Machine learning can also be used in co-ordination with technologies such as Internet of Things to detect diseases and hazardous situations without human intervention [10]. However, there is still problem of consistency and standardization in healthcare making it difficult to develop generalized machine learning algorithms [6].

D. Diagnosing heart diseases with machine learning

There has been a lot of research studying the role of machine learning algorithms in predicting heart failure or heart related diseases. [5] reviewed various machine learning models on the case of predicting heart diseases and examined the relationship between various model attributes to its accuracy. [11] analysed various KNN methods in detection of heart diseases. Adaptive KNN was found to produce the highest positive metrics suggesting its effectiveness in predicting cardiovascular disorders. [4] performed a comparative analysis to find the best algorithm for classification problem in heart failure prediction. The study found that the most effective algorithms were Naïve Bayes, Support Vector Machine and Random Forest algorithm. [6] presents the possible use of machine learning in medical decision support to diagnose various heart related diseases. Different datasets have been used to cover different category of heart diseases and the effectiveness of each algorithm has been tested on those datasets. The research concluded that models

may perform well in one case but perform worse on some other case. The research further indicates that these models face a problem of overfitting and are struggle with large unstructured data sets. [12]–[14] performed similar comparisons using the UCI dataset, all suggesting different model as having better effectiveness.

There are similar research using neural networks to predict heart diseases. [15] used various machine learning methods alongside ANN to predict the probability of heart diseases in an individual. [16] similarly compared machine learning models and ANNs and found ANNs to have the highest accuracy for the dataset. [17] proposed a Federated Learning (FL) based CNN model that can work with unstructured data from different sources. It aims to tackle medical data being unstructured due to privacy of patients. They achieved an accuracy of 95% for the dataset used in the study using this technique. [18] compared various deep learning algorithms with the UCI heart dataset. CNN, Long Short-Term Memory (LSTM), Recurrent Neural Network (RNN), DenseNet and Bi-LSTM models were compared, and they found that CNN had the best performance for the dataset.

III. METHODOLOGY

A. Macro methodology

When working with reports dealing with machine learning, a standard methodology can greatly improve the accuracy and clarity of the reports [19]. Various methods have been proposed to standardize the process, including KDD (Knowledge Discovery in Databases), CRISP-DM (Cross-industry Standard Process for Data Mining) and SEMMA (Sample, Explore, Modify, Model and Assess) [20]. In this report we use CRISP-DM as it is suited for data analytics and mining projects [19]. The following steps are involved in the process:

- 1) **Business Understanding:** This step involves setting the objectives of the research and expected goals.
- 2) **Data Understanding:** This step involves finding the correct dataset, exploring it, and finally checking the quality to meet standards.
- 3) **Data Preparation:** In this step the data is cleaned, and relevant rows and columns are selected for analysis.
- 4) **Modelling:** In this step the models for the analysis are defined using the dataset.
- 5) **Evaluation:** This step assesses whether the model met the research/business objectives correctly.
- 6) **Deployment:** If the evaluation

B. Micro Methodology

This section talks about the data and the methods used to fulfil the research objectives.

a) *Data:* For the analysis we are using the Heart Disease Health Indicators Dataset that is available on Kaggle. This dataset is based on the survey responses collected by Centres for Disease Control and Prevention (CDC) for their Behavioural Risk Factor Surveillance System (BRFSS) survey. The survey collects responses from over 400,000 Americans on health-related questions. The response was then compiled

into BRFSS 2015 dataset. The author then further cleaned and consolidated the previous dataset to create this one. It has 253,680 entries and 22 columns with one target column for study [21].

TABLE I: Attributes in the dataset

Attribute	Description	Value Range
HeartDiseaseorAttack	Heart disease or attack presence	0.0 - 1.0
HighBP	High blood pressure presence	0.0 - 1.0
HighChol	High cholesterol presence	0.0 - 1.0
CholCheck	Cholesterol check undergone	0.0 - 1.0
BMI	Body Mass Index	12.0 - 98.0
Smoker	Smoking status	0.0 - 1.0
Stroke	Stroke occurrence	0.0 - 1.0
Diabetes	Diabetes status	0.0 - 2.0
PhysActivity	Physical activity engagement	0.0 - 1.0
Fruits	Fruit consumption	0.0 - 1.0
Veggies	Vegetable consumption	0.0 - 1.0
HvyAlcoholConsump	Heavy alcohol consumption	0.0 - 1.0
AnyHealthcare	Access to healthcare	0.0 - 1.0
NoDocbcCost	Doctor visit affordability	0.0 - 1.0
GenHlth	General health status	1.0 - 5.0
MentHlth	Mental health status	0.0 - 30.0
PhysHlth	Physical health status	0.0 - 30.0
DiffWalk	Difficulty walking	0.0 - 1.0
Sex	Sex of the individual	0.0 - 1.0
Age	Age category	1.0 - 13.0
Education	Education level	1.0 - 6.0
Income	Income level	1.0 - 8.0

b) Machine Learning Algorithms:

- **Decision Trees:** Decision trees are a supervised machine learning method for classification and regression. It uses a tree like structure to distribute data. It uses maximum information gain or other constraints to split the root into sub-trees recursively, until a leaf node is reached. It can be used on both categorical and numerical data, but it might suffer from overfitting if there are too many branches [4].
- **Random Forest:** Random forest uses decision trees for training and prediction. It is a supervised learning technique that uses ensemble learning. Every record has its own decision tree, and the prediction is done by combining all the decision trees. For better accuracy, it is suggested to use it with a large dataset [4].
- **Logistic Regression:** Logistic regression is another classification algorithm. Its output lies in the range of 0 to 1, based on maximum likelihood estimation. The output is a probability, so a threshold is used to separate between binary classification (Battula et al., 2021). Logistic regression has disadvantages when working with small dataset where the quantity of data is lower than its features [4].
- **Naïve Bayes:** Naïve Bayes algorithm is a classifier that calculates a set of probability by counting the frequency and combinations of values in given data set. It is characterized as being Naïve because of its assumption of conditional independence for all variables. It is based on Bayes' theorem [22].
- **K-nearest Neighbours:** K-nearest Neighbours (KNN) is a type of classification algorithm that categorizes data points based on their distance to other points in training data set. KNN are sensitive to irrelevant attributes that obscure important attributes [23].
- **Gradient Boosting:** Gradient boosting algorithm has been implemented using the scikit-learn library.

c) *Deep Learning Algorithms:*

- **Artificial Neural Networks:** Artificial Neural Networks or ANNs are a type of deep learning algorithm that use three layers to classify or predict some information. It consists of input layer, hidden layer, and the output layer. The input layer is linked to the input structure of the data. Hidden layer consists of different interconnected nodes that have various weights between connection, like a human brain. The output layer gives the output calculated by the hidden layer [24]. The algorithm uses an error correction mechanism such as backward-propagation or root mean square error to adjust the weights to improve its accuracy [15].
 - **Convolutional Neural Network:** Convolutional Neural Networks (CNN) are a type of deep learning algorithm which is commonly used for classification and prediction. It consists of convolutional layers, pooling layers, and fully connected layers are all hidden layers of CNN [17].
- d) *Machine Learning Libraries:*
- **scikit-learn:** scikit-learn has been used for majority of the machine learning classification algorithms. Also, various metrics have been used from its library for model evaluation.
 - **Matplotlib:** Matplotlib has been used for plotting the graphs obtained in the analysis.
 - **Pandas:** Pandas has been used for data pre-processing and handling the dataset.
 - **TensorFlow:** TensorFlow is used alongside Keras for building deep learning models.
 - **Imbalanced-learn:** This library has been used to remove imbalances in the data.

IV. FINDINGS AND DISCUSSION

The above mentioned algorithms and models were implemented and evaluated to obtain the following results. Each section describes the findings of a particular evaluation concern.

A. Data Pre-processing

Starting with the dataset, the structure of the data was explored and then pre-processed to use it in the next step. The data initially consisted of 253,680 entries and 22 columns in total. Out of the 22 columns only 12 features were selected to limit the evaluation into the scope of the research. Features dealing with nutrition, lifestyle and physical health were selected whereas features such as Age, Education, Income and so on were discarded. The feature HeartDiseaseorAttack was selected as the target column for this study.

Also, to deal with the class imbalance in the data Imbalanced-learn library was used to create a balance between the two target classes. Using a ratio of 1:1, both classes were balanced to have 23,893 occurrences each. The occurrences dropped significantly from the initial count but this was done to reduce bias towards one target class. The independent features were then scaled using standard scaler method provided by scikit-learn. Finally, the data was split into testing and

training. 80% of the data was used towards training whereas 20% of the data was used for testing, resulting in a test size of 0.2.



Fig. 1: Count of target class 0 and target class 1 in the dataset

B. Machine Learning

Six machine learning models were tested on the pre-processed data with similar parameters. Table II shows the results obtained from the evaluation.

TABLE II: Evaluation metrics of various machine learning models

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Decision Tree	0.6546	0.6748	0.6077	0.6395	0.655
Random Forest	0.681	0.6816	0.6891	0.6853	0.6809
Logistic Regression	0.7205	0.7229	0.7225	0.7227	0.7205
Naive Bayes	0.6933	0.7203	0.6403	0.6779	0.6938
K-Nearest Neighbors	0.6884	0.6853	0.7061	0.6956	0.6883
Gradient Boosting	0.7266	0.7176	0.7547	0.7357	0.7264

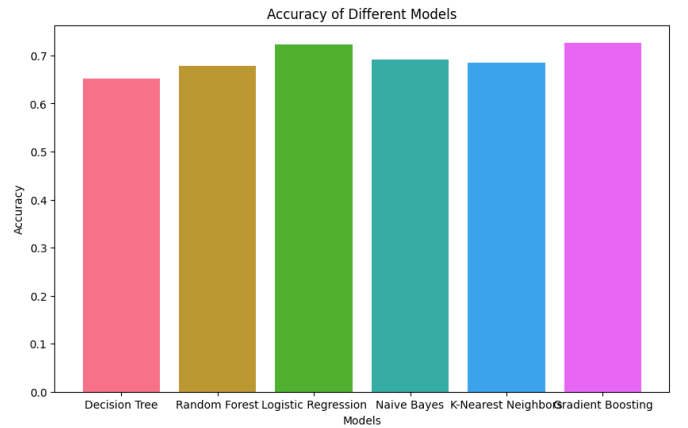


Fig. 2: Accuracy comparison of the machine learning models

From the table and figure we can see that Gradient Boosting model performed the best with a accuracy of 72.66% followed by Logistic Regression. Decision tree on the other hand was the worst performing model with an accuracy of 65

Next to compare the predictive performance of the models three scenarios were created, an unhealthy individual, an healthy individual and an individual with mixed features. The

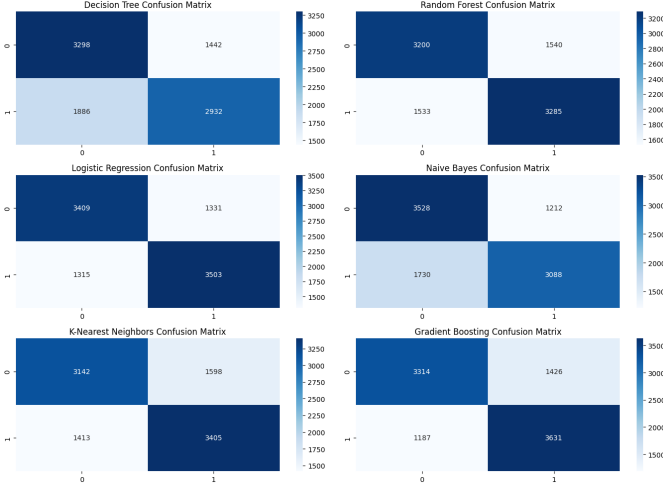


Fig. 3: Confusion Matrix for machine learning models

responses are synthetically generated to observe the predictive performance of the models in new data.

TABLE III: Synthetic survey scenarios for testing prediction

Attribute	Unhealthy	Healthy	Mixed
Fruits	0	1	0
PhysActivity	0	1	1
Veggies	0	1	1
PhysHlth	30	0	15
DiffWalk	1	0	1
HvyAlcoholConsump	1	0	0
HighBP	1	0	1
HighChol	1	0	1
Diabetes	1	0	0
GenHlth	5	1	3
BMI	40	22	30

Table IV shows the predicted probability of getting a heart disease by the models. All the models have predicted a different probability for the three individuals. For example, decision tree predicted that the Unhealthy individual will surely get a heart attack because it predicted a probability of 1. Meanwhile Random Forest and Logistic Regression predicted a probability of 87%. Naive Bayes however predicted that the individual with mixed attributes has 96% probability of getting a heart disease which varies significantly from all other models.

C. Deep learning

For deep learning a Feed-forward neural network (FF-NN) and Convolutional Neural Network (CNN) were implemented for prediction.

For FF-NN we used a model with three layers. The input layer is a dense layer with 32 units. The input layer has size equal to the attributes in training data. Rectified Linear Unit (ReLU) is used as the activation function. The hidden layer consists of 16 units with same ReLU activation function. Output layer has a single unit as it is predicting a binary output. Sigmoid function is used as activation function here to capture the output between 0 and 1 and a threshold of 0.5 is used for prediction. The model is trained for 20 iterations with a batch size of 32 and a validation split of 20%. The results of the training are given presented in the figures 4 and 5.

Model	Individual	Probability
Decision Tree	Unhealthy Individual	1.00
	Healthy Individual	0.05
	Mixed Individual	0.00
Random Forest	Unhealthy Individual	0.87
	Healthy Individual	0.05
	Mixed Individual	0.53
Logistic Regression	Unhealthy Individual	0.87
	Healthy Individual	0.11
	Mixed Individual	0.74
Naive Bayes	Unhealthy Individual	0.99
	Healthy Individual	0.00
	Mixed Individual	0.96
K-Nearest Neighbors	Unhealthy Individual	0.60
	Healthy Individual	0.20
	Mixed Individual	0.60
Gradient Boosting	Unhealthy Individual	0.88
	Healthy Individual	0.08
	Mixed Individual	0.74

TABLE IV: Predicted probabilities for three survey response scenarios

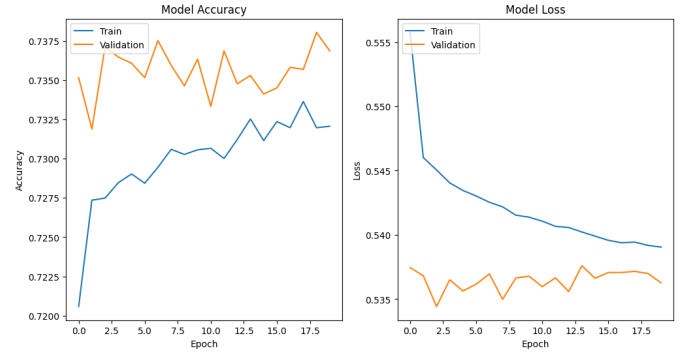


Fig. 4: Model accuracy and loss for the Feed-forward Neural Network over 20 iterations

CNN model on the other hand was implemented using 1-dimensional (1D) layers. The input layer is a 1D convolutional layer with 32 filters and kernel size of 3. ReLU is used as the activation layer. The layer is followed by MaxPooling layer of pool size 2. Further, the hidden layer is another 1D convolutional layer with 64 filters and kernel size 3. A MaxPooling layer follows with pool size 2. 2 dimensional output of previous layer is flattened to 1D by using a flatten layer. The flattened output is fed to a dense layer with 64 units and finally to the output layer with 1 unit for binary classification. Sigmoid is used as the activation function with 0.5 as the threshold. For optimization, Adam optimizer was used and Binary Cross-entropy was used for loss function. The results of the training are presented in the figures 6 and 7.

Metric	FF-NN	CNN
Accuracy	0.7263	0.7234
Precision	0.7232	0.7164
Recall	0.7403	0.7470
F1 Score	0.7317	0.7314
ROC-AUC	0.7262	0.7920

TABLE V: Performance comparison of Feed-forward Neural-Network (FF-NN) and Convolutional Neural Network (CNN)

Table V compares the accuracies of both the deep learning

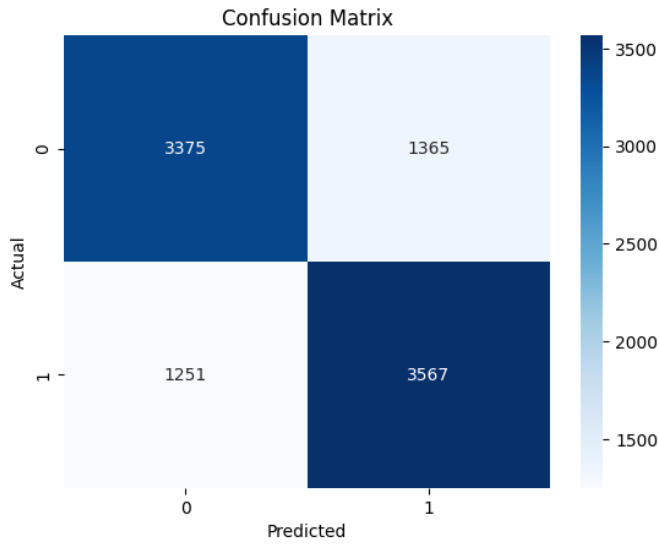


Fig. 5: Confusion matrix for FF-NN

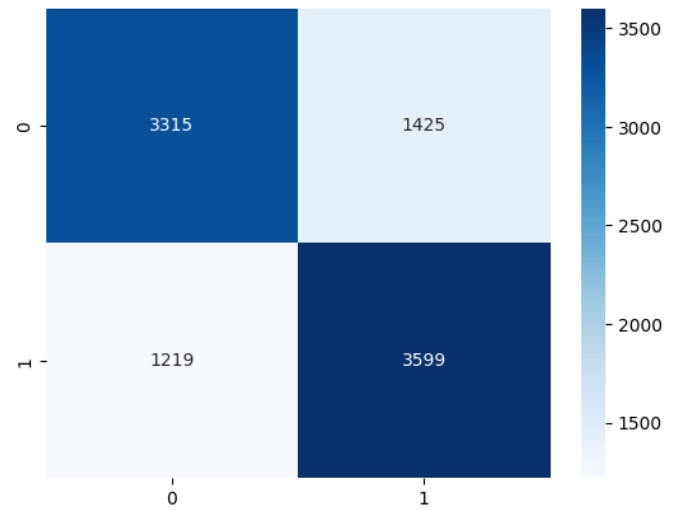


Fig. 7: Confusion matrix for CNN



Fig. 6: Model accuracy and loss for the CNN over 20 iterations

models From the table it can be observed that both the models perform very similarly for the given dataset. The accuracy obtained is very comparable to ones obtained by machine learning model Gradient Boost and Logistic Regression.

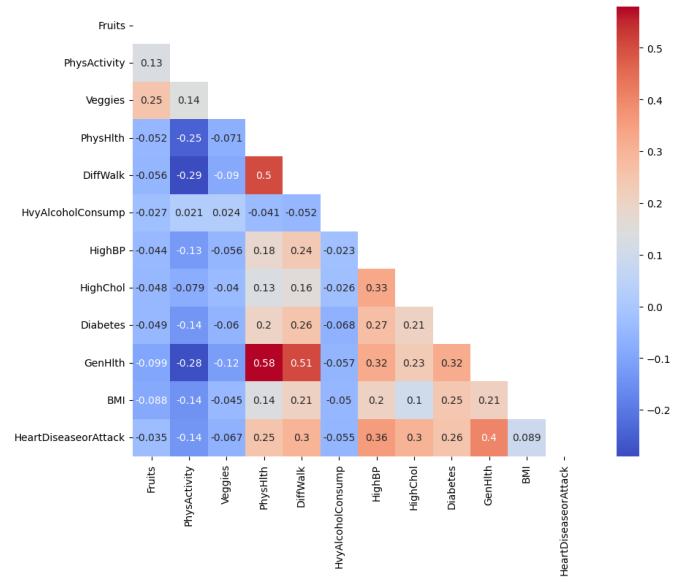


Fig. 8: Correlation matrix for all features

D. Correlation

Figure 8 shows the correlation matrix between the various health indicators and the target variable that is the likelihood of heart disease. It is observed that “GenHlth” (General Health) is moderately correlated with the target, suggesting that worse general health is associated with greater risk of heart disease. Similarly, other features like Diabetes, HighBP (high blood pressure), and HighChol (High cholesterol) have positive correlation with the target, aligning with the literature reviewed in previous sections. In contrast, behavioral features like PhysActivity (physical activity) and Fruits(eating fruits) exhibit a negative correlation suggesting that healthier lifestyles choices might be associated with lowered risk of heart diseases. With this analysis, the key indicators for heart health can be identified and validated with existing literature.

V. CONCLUSION AND RECOMMENDATION

This report looked into the prediction of heart disease by using various machine learning and deep learning techniques. Various machine learning and deep learning models were evaluated and it was found that Gradient Boost performed the best with the dataset. Logistic Regression, FF-NN and CNN all had similar performance to Gradient Boost. The report used a survey based dataset to perform the evaluations and it was evident that after appropriate pre-processing, such dataset can be used to perform predictions. Through correlation testing, it was found factors such as physical activity, nutrition have a positive impact on heart health while factors such as high blood pressure and diabetes negatively impact an individual's heart health. This findings are aligned with what is proven in medical literature.

A. Recommendation

Based on the results of the analysis, it is evident that most of the models had similar performance for predicting heart disease using the BRFS dataset. So, it is recommended to use simpler models that are efficient instead of using computationally expensive models like CNN for this scenario. Regarding the use of machine learning models in prediction of heart disease, it can be seen that various models predicted different probabilities of getting heart disease based on an synthetic scenario. It is recommended to follow the prediction of model having the best performance while validating the result with a medical professional as the models itself might not be able to capture all the aspects of heart disease through survey response alone. A larger dataset is recommended to remove the problem of over-fitting even if it results in reduced accuracy of the models. This can help the model cover various scenarios that may not be captured by a small dataset. Since the structure of dataset can influence the model it is recommended to evaluate the models on a case by case basis rather than generalizing the performance. The results of this evaluation can be used as reference for a dataset with similar structure. Finally, it is recommended to follow the results of correlation testing to identify the most influential factors in heart health to provide a basis for change in lifestyle choices.

B. Future Work

The evaluation of the models using this dataset can further be improved by using all the parameters of the dataset and also by using all the rows by through a different balancing method. The study does not use full features of the models, so various parameter tuning techniques can be used to improve the accuracy of the models. Instead of using synthetic scenarios, real life survey can be taken to evaluate the performance of the models and to validate the results. Further, more research can be done towards standardizing the data from various sources so that an generalized model can be built for prediction of heart diseases. Such a step required evaluating what the best data structure will be for the models available today, or changing the models to best suit the data that healthcare generates.

REFERENCES

- [1] S. Coughlin, D. Roberts, K. O'Neill, and P. Brooks, "Looking to tomorrow's healthcare today: a participatory health perspective," *Internal Medicine Journal*, vol. 48, pp. 92–96, 1 2018.
- [2] W. Pawłowski, K. Goniewicz, M. Goniewicz, and D. Lasota, "Lifestyle and wellness-effects on cardiovascular disease risk factors," pp. 228–233, 2018. [Online]. Available: www.monz.pl/PRACAORYGINALNA
- [3] D. Bloom, E. Cafiero, E. Jané-Llopis, S. Abrahams-Gessel, L. Bloom, S. Fathima, A. Feigl, T. Gaziano, M. Mowafi, A. Pandya, K. Prettnier, L. Rosenberg, B. Seligman, A. Stein, and C. Weinstein, "The global economic burden of non-communicable diseases," 2011. [Online]. Available: www.weforum.org/EconomicsOfNCD
- [4] C. D. Silva and P. Kumarawadu, "Performance analysis of machine learning classification algorithms in the case of heart failure prediction." Institute of Electrical and Electronics Engineers Inc., 2022, pp. 1160–1165.
- [5] N. Tyagi and P. Jain, "A review of machine learning algorithms for predicting heart disease." Institute of Electrical and Electronics Engineers (IEEE), 4 2024, pp. 961–965.
- [6] S. Ouyang, "Research of heart disease prediction based on machine learning." Institute of Electrical and Electronics Engineers Inc., 2022, pp. 315–319.

- [7] D. C. Mariachiara, H. Bixby, T. Gaziano, L. Hadeed, C. Kabudula, D. V. McGhie, J. Mwangi, B. Pervan, P. Perel, D. Piñeiro, S. Taylor, and F. Pinto, "World heart report 2023," 2023.
- [8] P. Ponikowski, S. D. Anker, K. F. AlHabib, M. R. Cowie, T. L. Force, S. Hu, T. Jaarsma, H. Krum, V. Rastogi, L. E. Rohde, U. C. Samal, H. Shimokawa, B. B. Siswanto, K. Sliwa, and G. Filippatos, "Heart failure: preventing disease and death worldwide," pp. 4–25, 9 2014.
- [9] B. Schrader, A. M. Bünker, C. Conradi, S. Lüders, B. Vaske, M. Kozi-olek, H. Haller, A. Elsässer, and J. Schrader, "Regular exercise is associated with a more favorable cardiovascular risk profile, better quality of life, less depression and less psychological stress," *International Journal of General Medicine*, vol. 15, pp. 545–554, 2022.
- [10] E. S. Tumpa and K. Dey, "A review on applications of machine learning in healthcare." Institute of Electrical and Electronics Engineers Inc., 2022, pp. 1388–1392.
- [11] L. M. Kunjachen and R. Kavitha, "Comparative study on detection of heart disease using knn algorithm." Institute of Electrical and Electronics Engineers Inc., 2022, pp. 14–19.
- [12] K. Battula, R. Durgadinesh, K. Suryapratap, and G. Vinaykumar, "Use of machine learning techniques in the prediction of heart disease." Institute of Electrical and Electronics Engineers Inc., 10 2021.
- [13] B. Kumari, S. Tripathy, and S. Sahoo, "Machine learning algorithms for detecting the existence of heart disease in patients: a comparative analysis." Institute of Electrical and Electronics Engineers Inc., 2022.
- [14] H. Yu, "Analysis and prediction of heart disease based on machine learning algorithms." Institute of Electrical and Electronics Engineers Inc., 2023, pp. 1418–1423.
- [15] M. Priya, A. Anitha, M. K. Nallakaruppan, J. Raju, R. Raghavan, and D. Srivastava, "Heart disease prediction using machine learning algorithms." Institute of Electrical and Electronics Engineers Inc., 2023.
- [16] M. S. Gangadhar, K. V. S. Sai, S. H. S. Kumar, K. A. Kumar, M. Kavitha, and S. S. Aravinth, "Machine learning and deep learning techniques on accurate risk prediction of coronary heart disease." Institute of Electrical and Electronics Engineers Inc., 2023, pp. 227–232.
- [17] P. Sharma and S. Sharma, "An effective fl-cnn based data securing model for heart disease prediction." Institute of Electrical and Electronics Engineers Inc., 2023, pp. 1862–1866.
- [18] A. Jaiswal, M. Singh, and N. Sachdeva, "Empirical analysis of heart disease prediction using deep learning." Institute of Electrical and Electronics Engineers Inc., 2023.
- [19] C. Schröder, F. Kruse, and J. M. Gómez, "A systematic literature review on applying crisp-dm process model," vol. 181. Elsevier B.V., 2021, pp. 526–534.
- [20] U. Shafique and H. Qaiser, "A comparative study of data mining process models (kdd, crisp-dm and semma)," pp. 217–222, 2014. [Online]. Available: <http://www.ijisr-issr-journals.org/>
- [21] A. Teboul, "Heart disease health indicators dataset," 2022. [Online]. Available: <https://www.kaggle.com/datasets/alextteboul/heart-disease-health-indicators-dataset/data>
- [22] M. M. Saritas and A. Yasar, "Performance analysis of ann and naive bayes classification algorithm for data classification," *Original Research Paper International Journal of Intelligent Systems and Applications in Engineering IJISAE*, vol. 7, pp. 88–91, 2019. [Online]. Available: www.ijisae.org
- [23] MathWorks, "Choosing the best machine learning classification model and avoiding overfitting." [Online]. Available: <https://uk.mathworks.com/campaigns/offers/next/choosing-the-best-machine-learning-classification-model-and-avoiding-overfitting.html>
- [24] C.-C. Peng, C.-W. Huang, and Y.-C. Lai, *Heart Disease Prediction Using Artificial Neural Networks: A Survey*, 2020.

APPENDIX

A. Dataset and Notebook

The dataset and Google Colab notebook used in this study can be accessed at the following drive folder:

- Resources: Resource Link